Dylan Thornton

Assignment #5 (30 points)

Due on Monday 05/12/2025 at 11:59pm- Submit on Camino

Make sure to calculate the weighted impurity indices for question 2. The weighted impurity indices should be the base of making decision on the best feature for splitting in question 3.

Useful Formulas

Here are some formulas and values that may be useful for completing this assignment:

Gini Index:

$$Gini(t) = 1 - \sum_{i=0}^{c-1} [p(i|t)]^2$$

Entropy:

$$Entropy(t) = -\sum_{i=0}^{c-1} p(i|t) \log_2 p(i|t)$$

Classification Error:

$$Classification\ Error(t) = 1 - \max_i [p(i|t)]$$

Where $p(i|t)$ is the probability of class i at node t.

Assignment Questions

You are provided with a dataset of animals and their attributes, as shown below. Your task is to determine the best attribute to split the dataset using the **Gini Index**, **Entropy**, and **Classification Error** measures of impurity. You will calculate these measures for three attributes: **Can Fly**, **Live in Water**, and **Have Legs**. The purpose is to find the best attribute for splitting.

Question 1. Class Distribution for Each Attribute (4 points):

For the three attributes **Can Fly**, **Live in Water**, and **Have Legs**), count how many animals belong to each class (Mammal or Non-mammal).

Question 2. Impurity Calculation (24 points):

For each attribute, calculate the **Gini Index**, **Entropy**, and **Classification Error** based on the class distribution.

- Provide the formulas used for each impurity measure and show all steps for calculating them.

Question 3. Best Attribute for Splitting (2 points):

Compare the impurity measures for all attributes and also the measures for **Give Birth** that was included in the lecture notes. Determine which attribute offers the best split by minimizing the impurity. Justify your selection with calculations and an explanation of why this attribute is the best for splitting the dataset.

Dataset:

total=20

Name	Give Birth	Can Fly	Live in Water	Have Legs	Class
Human	Yes	No	No	Yes	Mammal
Python	No	No	No	No	Non-mammal
Salmon	No	No	Yes	No	Non-mammal
Whale	Yes	No	Yes	No	Mammal
Frog	No	No	Sometimes	Yes	Non-mammal
Komodo	No	No	No	Yes	Non-mammal
Bat	Yes	Yes	No	Yes	Mammal
Pigeon	No	Yes	No	Yes	Non-mammal
Cat	Yes	No	No	Yes	Mammal
Leopard Shark	Yes	No	Yes	No	Non-mammal
Turtle	No	No	Sometimes	Yes	Non-mammal
Penguin	No	No	Sometimes	Yes	Non-mammal
Porcupine	Yes	No	No	Yes	Mammal
Eel	No	No	Yes	No	Non-mammal
Salamander	No	No	Sometimes	Yes	Non-mammal
Gila Monster	No	No	No	Yes	Non-mammal
Platypus	No	No	No	Yes	Mammal
Owl	No	Yes	No	Yes	Non-mammal
Dolphin	Yes	No	Yes	No	Mammal
Eagle	No	Yes	No	Yes	Non-mammal

Question 1. Class Distribution for Each Attribute (4 points):

For the three attributes **Can Fly**, **Live in Water**, and **Have Legs**), count how many animals belong to each class (Mammal or Non-mammal).

	Can Fly		Live in Water			Have legs	
	yes	no	yes	sometimes	no	yes	no
Mamamal	1	6	2	0	5	5	2
Not Mamamal	3	10	3	4	6	9	4

Question 2. Impurity Calculation (24 points):

For each attribute, calculate the **Gini Index**, **Entropy**, and **Classification Error** based on the class distribution.

- Provide the formulas used for each impurity measure and show all steps for calculating them.

Can Fly

16 cannot

$1-((\frac{6}{16})^2+(\frac{10}{16})^2)=.46875$

4 can

$1-((\frac{1}{4})^2+(\frac{3}{4})^2)=.375$

$.46875(.8)+.375(.2)=.45$

$\boxed{\text{Gini}=.45}$

Live in Water

11 can't

$1-((\frac{5}{11})^2+(\frac{6}{11})^2)=.496$

5 can

$1-((\frac{2}{5})^2+(\frac{3}{5})^2)=.48$

4 can

$1-((\frac{4}{4})^2+(\frac{0}{4})^2)=0$

$\frac{11}{20}(.496)+\frac{5}{20}(.48)+$

$\frac{4}{20}(0)=\boxed{.393=\text{Gini}}$

Have Legs

6 don't

$1-((\frac{2}{6})^2+(\frac{4}{6})^2)=\frac{4}{9}$

14 do

$1-((\frac{5}{14})^2+(\frac{9}{14})^2)=.459$

$\frac{6}{20}(\frac{4}{9})+.459(\frac{14}{20})=.455$

$\boxed{.455=\text{gini}}$

Class error

$\frac{6}{16}(.8)+\frac{1}{4}(.2)=\boxed{.35}$

Class error

$\frac{2}{5}(\frac{5}{20})+\frac{0}{4}(\frac{4}{20})+\frac{5}{11}(\frac{11}{20})=\boxed{.35}$

Class error

$\frac{5}{14}(\frac{14}{20})+\frac{2}{6}(\frac{6}{20})=\boxed{.35}$

Entropy

$-(\frac{1}{4}\log_2(\frac{1}{4})+\frac{3}{4}\log_2(\frac{3}{4}))\frac{4}{20}$

$+-(\frac{6}{16}\log_2\frac{6}{16}+\frac{10}{16}\log_2\frac{10}{16})\frac{16}{20}$

$=.162+.7635=\boxed{.926}$

Entropy

$-(\frac{2}{5}\log_2(\frac{2}{5})+\frac{3}{5}\log_2(\frac{3}{5}))\frac{5}{20}$

$+-(\frac{5}{11}\log_2(\frac{5}{11})+\frac{6}{11}\log_2(\frac{6}{11}))\frac{11}{20}$

$+-(\frac{4}{4}\log_2(0)+1\log_2(1))\frac{4}{20}=$

$.547+.243=\boxed{.789}$

Entropy

$-(\frac{5}{14}\log_2(\frac{5}{14})+\frac{9}{14}\log_2(\frac{9}{14}))\frac{14}{20}$

$+-(\frac{2}{6}\log_2(\frac{2}{6})+\frac{4}{6}\log_2(\frac{4}{6}))\frac{6}{20}=$

$.658+.275=\boxed{.934}$

Question 3. Best Attribute for Splitting (2 points):

Compare the impurity measures for all attributes and also the measures for **Give Birth** that was included in the lecture notes. Determine which attribute offers the best split by minimizing the impurity. Justify your selection with calculations and an explanation of why this attribute is the best for splitting the dataset.

Give birth:

Gini = .255

entropy = .573

class error = .15

It's clearly Give Birth as GB has the lowest impurity across every metric.

.255 < .343 < .45 < .455

.15 < .35 = .35 = .35

.573 < .784 < .926 < .934

Based on these ~~#s~~, I'd rank the attributes as follows for best split

1. Gives Birth
2. Lives in water
3. Can Fly
4. Have legs

Again this is simply based on which vals are lowest.